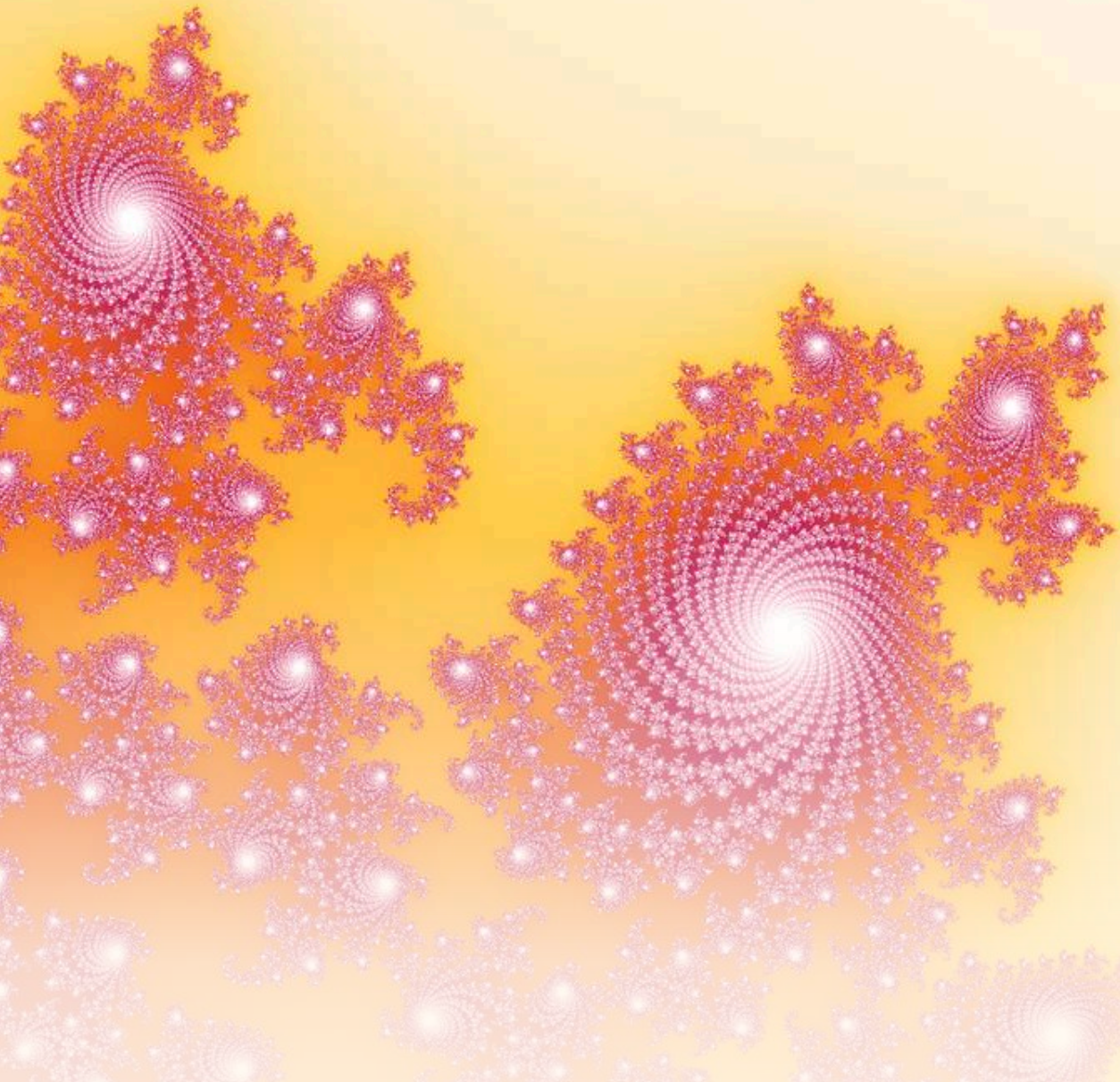


YEAR 9 STANDARD → YEAR 10 STANDARD

Summer Maths

Ready for Year 10

Six weeks. Twenty questions a week. Every Year 9 skill that Year 10 Standard builds on – mixed together so it sticks.

Week 1 | Week 2 | Week 3 | Week 4 | Week 5 | Week 6

NAME _____ CLASS _____

How it works

One page a week

Each week has **20 questions** and a **bonus**. Set aside about **40 minutes** – one sitting or a few short ones. Questions start easier and get a little harder through the page.

The same pattern every week

Question 1 is always fractions and integers, question 11 is always Pythagoras, and so on. If the same number keeps going wrong, that is the skill to revise before Year 10.

1	Fractions & integers	11	Pythagoras
2	Percentages	12	Trigonometry
3	Sig. figs & sci. notation	13	Area & surface area
4	Index laws	14	Volume
5	Zero & negative indices	15	Gradient & midpoint
6	Simplify & substitute	16	Straight lines
7	Expand	17	Data
8	Factorise	18	Probability
9	Equations	19	Earning money
10	Formulas & word problems	20	Problem solving

Why Julia? Each banner is part of the same *Julia set*: pick a number c, then repeat $z \to z^2 + c$ for every point. Points that stay small are coloured in. Every booklet in the series zooms into a different corner of the same set – this one zooms in on a cluster of small spirals.

NSW syllabus (NESA Mathematics K–10, 2022). Revises MA5-FIN-C-01, MA5-ALG-C-01, MA5-IND-C-01, MA5-EQU-C-01, MA5-LIN-C-01, MA5-LIN-C-02, MA5-MAG-C-01, MA5-TRG-C-01, MA5-ARE-C-01, MA5-VOL-C-01, MA5-GEO-C-01, MA5-DAT-C-01, MA5-PRO-C-01, with MA4-PYT-C-01. ★ Year 10 preview: MA5-FIN-C-02, MA5-TRG-C-02, MA5-ARE-P-01, MA5-VOL-P-01, MA5-ALG-P-02, MA5-EQU-P-01, MA5-EQU-P-02, MA5-LIN-C-02 (perpendicular lines).

Rules of the game

- **No calculator.** Leave answers exact (e.g. 48π) unless asked to round. Trig questions give you the value you need.
- **Show working** in the space under each question.
- **Mark it yourself** using the answers at the back (pages 10–12). Each answer has a line of working to help you find a mistake.
- **Fix it.** Redo anything you got wrong a day or two later, without looking.
- ★ questions preview Year 10 Standard work. Have a go – they are the ones that will help most in Term 1.
- **Stuck?** The hints page (page 9) has the key facts and formulas.

My summer tracker

WEEK	DATE	SCORE	DONE	BONUS	REVISE (Q NUMBERS)
1 Week 1		/ 20	☐	☐	
2 Week 2		/ 20	☐	☐	
3 Week 3		/ 20	☐	☐	
4 Week 4		/ 20	☐	☐	
5 Week 5		/ 20	☐	☐	
6 Week 6		/ 20	☐	☐	

First up in Year 10 Standard: interest and depreciation, graphing lines, surface area and volume, algebra and equations. Every one of them starts from the skills in these six pages.

WEEK 1

Summer Maths

Ready for Year 10

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices · Algebra · Equations · Measurement & trig · Coordinate geometry · Data & chance · Money & context

★ Year 10 preview (Stage 5) · No calculator

1 FRACTIONS & INTEGERS

Evaluate $-12 + 3 \times (-4)$

2 PERCENTAGES

Increase \$250 by 12%.

3 SIG. FIGS & SCI. NOTATION

Round 0.05348 to 2 significant figures.

4 INDEX LAWS

Simplify $a^5 \times a^3$

5 ZERO & NEGATIVE INDICES

Evaluate $5^0 + 3^{-1}$

6 SIMPLIFY & SUBSTITUTE

Simplify $4x - 7y - 2x + 3y$

7 EXPAND

Expand and simplify $3(2x - 5) + 4x$

8 FACTORISE

Factorise $9a^2 - 6a$

9 EQUATIONS

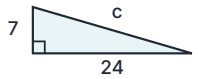
Solve $4x - 9 = 2x + 5$

10 FORMULAS & WORD PROBLEMS

$s = ut + \frac{1}{2}at^2$. Find s when $u = 3$, $t = 4$ and $a = 2$.

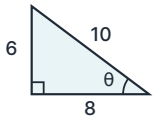
11 PYTHAGORAS

Find c. (cm)



12 TRIGONOMETRY

Write $\sin\theta$ as a fraction in simplest form.



13 AREA & SURFACE AREA

Find the surface area of a cube with edges 5 cm long.

14 VOLUME

Find the volume of a cylinder with radius 4 cm and height 5 cm, in terms of π.

15 GRADIENT & MIDPOINT

Find the midpoint of the interval joining $(2,\ 7)$ and $(8,\ -1)$.

16 STRAIGHT LINES

State the gradient and y-intercept of $y = -2x + 5$.

17 DATA

Find the mean of $12,\ 15,\ 9,\ 18,\ 16$.

18 PROBABILITY

Two coins are tossed. Find $P(\text{at least one head})$.

19 EARNING MONEY

Liam earns \$28 an hour. Find his pay for a 35-hour week.

20 PROBLEM SOLVING

A phone plan costs \$45 a month plus 10 cents per text. Find the cost of a month with 120 texts.

BONUS The sum of three consecutive even numbers is 78. Find them.

WEEK **2** Summer Maths

Ready for Year 10

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices · Algebra · Equations · Measurement & trig · Coordinate geometry · Data & chance · Money & context

★ Year 10 preview (Stage 5) · No calculator

1 FRACTIONS & INTEGERS

Evaluate $\frac{2}{3} \times \frac{9}{10} - \frac{1}{5}$

2 PERCENTAGES

A jacket marked \$120 is reduced by 35%. Find the sale price.

3 SIG. FIGS & SCI. NOTATION

Write $4\,720\,000$ in scientific notation.

4 INDEX LAWS

Simplify $(2m^3)^2$

5 ZERO & NEGATIVE INDICES

Write 10^{-3} as a decimal.

6 SIMPLIFY & SUBSTITUTE

If $p = -2$ and $q = 3$, find the value of $p^2 - 4pq$.

7 EXPAND

Expand $(x + 6)(x - 2)$

8 FACTORISE

Factorise $14xy + 21y$

9 EQUATIONS

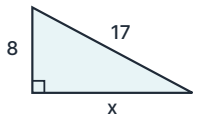
Solve $5(x + 2) = 3x + 18$

10 FORMULAS & WORD PROBLEMS

Rachel is 3 times as old as her brother. In 4 years their ages will add to 36. How old is her brother now?

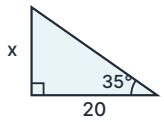
11 PYTHAGORAS

Find x. (cm)



12 TRIGONOMETRY

Use $\tan 35^\circ \approx 0.70$ to find x. (cm)



13 AREA & SURFACE AREA

Find the area of a quadrant (quarter circle) with radius 6 cm, in terms of π.

14 VOLUME

A triangular prism has a triangle with base 8 cm and height 6 cm as its cross-section, and length 15 cm. Find its volume.

15 GRADIENT & MIDPOINT

Find the gradient of the line through $(-1,\ 4)$ and $(3,\ -8)$.

16 STRAIGHT LINES

Find the equation of the line with gradient 3 that passes through $(0,\ -4)$.

17 DATA

Find the median of the data in the stem-and-leaf plot.

2	3 5 8
3	0 2 2 6
4	1 4 9

Key: 3 | 2 means 32

18 PROBABILITY

A card is drawn at random from cards numbered 1 to 12. Find $P(\text{factor of } 12)$.

19 EARNING MONEY

Sam's normal rate is \$30 an hour. He works 38 hours at the normal rate and 4 hours at time-and-a-half. Find his pay.

20 PROBLEM SOLVING

A map has a scale of $1 : 25\,000$. A walking track is 14 cm long on the map. Find its real length in km.

BONUS Show that $(n + 1)^2 - (n - 1)^2 = 4n$ for every number n.

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices · Algebra · Equations · Measurement & trig · Coordinate geometry · Data & chance · Money & context

★ Year 10 preview (Stage 5) · No calculator

1 FRACTIONS & INTEGERS

Evaluate $3\frac{3}{4} - 1\frac{5}{6}$

2 PERCENTAGES

Write \$45 as a percentage of \$360.

3 SIG. FIGS & SCI. NOTATION

Write 3.06×10^{-4} as a decimal.

4 INDEX LAWS

Simplify $12x^7 \div 4x^2$

5 ZERO & NEGATIVE INDICES

Evaluate $2^{-2} \times 8$

6 SIMPLIFY & SUBSTITUTE

Simplify $\frac{2a}{3} + \frac{a}{6}$

7 EXPAND

Expand $(2x + 1)(x + 4)$

8 FACTORISE ★

Factorise $x^2 + 7x + 12$

9 EQUATIONS

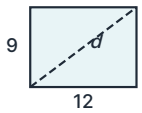
Solve $\frac{2x - 1}{3} = 5$

10 FORMULAS & WORD PROBLEMS

Use $F = \frac{9C}{5} + 32$ to find C when $F = 59$.

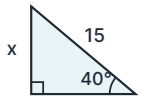
11 PYTHAGORAS

A rectangle is 9 cm by 12 cm. Find the length of its diagonal.



12 TRIGONOMETRY

Use $\sin 40° \approx 0.64$ to find x. (cm)



13 AREA & SURFACE AREA

Find the surface area of a rectangular prism 6 cm by 4 cm by 3 cm.

14 VOLUME

A tank is 2 m long, 1.5 m wide and 1 m high. How many litres does it hold? ($1\ \text{m}^3 = 1000$ L)

15 GRADIENT & MIDPOINT

Find the length of the interval joining $(1,\ 2)$ and $(7,\ 10)$.

16 STRAIGHT LINES

Does the point $(-2,\ 7)$ lie on the line $y = -3x + 1$?

17 DATA

Which of these is categorical data: shoe size, favourite beach, height, number of siblings?

18 PROBABILITY

A bag has 3 red and 2 blue marbles. One is drawn and replaced, then a second is drawn. Find $P(\text{both red})$.

19 EARNING MONEY

A real estate agent earns 2% commission on a \$650 000 sale. How much is the commission?

20 PROBLEM SOLVING

A hose delivers 12 L per minute. How long does it take to fill a 1500 L pool? Answer in hours and minutes.

BONUS

A rectangle has an area of 48 cm² and a perimeter of 28 cm. Find its length and width.

WEEK 4

Summer Maths

Ready for Year 10

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices · Algebra · Equations · Measurement & trig · Coordinate geometry · Data & chance · Money & context

★ Year 10 preview (Stage 5) · No calculator

1 FRACTIONS & INTEGERS

Evaluate $(-3)^3 - (-2)^2$

2 PERCENTAGES ★

$1000 is invested at 10% p.a. compound interest. Find its value after 2 years.

3 SIG. FIGS & SCI. NOTATION

Evaluate $(3 \times 10^4) \times (2 \times 10^3)$ in scientific notation.

4 INDEX LAWS

Simplify $(3a^2b)^3$

5 ZERO & NEGATIVE INDICES

Write $\frac{1}{49}$ as a power of 7.

6 SIMPLIFY & SUBSTITUTE

Simplify $\frac{3x}{4} - \frac{x}{2}$

7 EXPAND ★

Expand $(x-5)^2$

8 FACTORISE

Factorise $20m - 5m^2$

9 EQUATIONS

Solve $x^2 = 64$

10 FORMULAS & WORD PROBLEMS

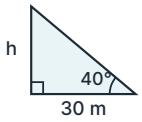
A taxi charges $3.60 plus $2.20 per km. A trip cost $47.60. How long was it?

11 PYTHAGORAS

A 13 m rope runs from the top of a vertical pole to a peg 5 m from its base. How tall is the pole?

12 TRIGONOMETRY ★

From 30 m away, the angle of elevation of the top of a tree is 40°. Use $\tan 40^\circ \approx 0.84$ to find its height.



13 AREA & SURFACE AREA

Find the surface area of a closed cylinder with radius 3 cm and height 5 cm, in terms of π.

14 VOLUME ★

Find the volume of a cone with radius 3 cm and height 10 cm, in terms of π.

15 GRADIENT & MIDPOINT

$M(4,\ 1)$ is the midpoint of AB, and A is $(1,\ -3)$. Find B.

16 STRAIGHT LINES

Find the equation of the line through $(0,\ 2)$ and $(3,\ 11)$.

17 DATA

Find the range and the mode of $15,\ 18,\ 12,\ 18,\ 21,\ 14,\ 18$.

18 PROBABILITY

A coin is tossed and a die is rolled. Find $P(\text{head and a } 6)$.

19 EARNING MONEY

Ella's salary is $78 000 a year. Find her fortnightly pay. (1 year = 26 fortnights)

20 PROBLEM SOLVING

A car travels 150 km in 1 hour 40 minutes. Find its average speed in km/h.

BONUS $2^x = 32$ and $3^y = \frac{1}{9}$. Find $x + y$.

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices · Algebra · Equations · Measurement & trig · Coordinate geometry · Data & chance · Money & context

★ Year 10 preview (Stage 5) · No calculator

1 FRACTIONS & INTEGERS

Evaluate $\frac{4}{5} \div \frac{8}{15}$

2 PERCENTAGES

A price rises from \$80 to \$92. Find the percentage increase.

3 SIG. FIGS & SCI. NOTATION

Round 7.0496 to 3 significant figures.

4 INDEX LAWS

Simplify $\frac{6x^5y^2}{2x^3y}$

5 ZERO & NEGATIVE INDICES

Evaluate $4^{-1} + 2^{-2}$

6 SIMPLIFY & SUBSTITUTE

If $m = \frac{1}{2}$ and $n = -4$, find the value of $6m - n^2$.

7 EXPAND

Expand and simplify $(x + 4)(x + 1) - 3x$

8 FACTORISE

Factorise $-8a - 12ab$

9 EQUATIONS

Solve $3(x - 4) - 2(x + 1) = 5$

10 FORMULAS & WORD PROBLEMS ★

Make r the subject of $C = 2\pi r$.

11 PYTHAGORAS

Is a triangle with sides 9 cm, 40 cm and 41 cm right-angled? Explain.

12 TRIGONOMETRY

In a right-angled triangle, $\tan\theta = \frac{5}{12}$. Find $\sin\theta$.

13 AREA & SURFACE AREA

A triangular prism has right-angled triangle ends with sides 3 cm, 4 cm and 5 cm, and length 10 cm. Find its surface area.

14 VOLUME

A cylinder has a volume of 72π cm³ and a height of 8 cm. Find its radius.

15 GRADIENT & MIDPOINT

Are $(0,\ 1)$, $(2,\ 5)$ and $(4,\ 9)$ on the same straight line? Explain.

16 STRAIGHT LINES ★

Find the gradient of a line perpendicular to $y = 4x - 7$.

17 DATA ★

Find the five-number summary of $3,\ 5,\ 6,\ 8,\ 9,\ 11,\ 14$.

18 PROBABILITY

Two dice are rolled. Find $P(\text{the sum is } 7)$.

19 EARNING MONEY

Kai is paid \$1.20 for each box he packs. How many boxes must he pack to earn \$480?

20 PROBLEM SOLVING ★

A \$30 000 car loses 20% of its value each year. Find its value after 2 years.

BONUS The diagonal of a square is 10 cm long. Find the area of the square.

WEEK 6

Summer Maths

Ready for Year 10

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices · Algebra · Equations · Measurement & trig · Coordinate geometry · Data & chance · Money & context

★ Year 10 preview (Stage 5) · No calculator

1 FRACTIONS & INTEGERS

Evaluate $-2\frac{1}{3} + 4\frac{1}{2}$

2 PERCENTAGES

After a 15% increase, the rent is \$460 a week. What was it before?

3 SIG. FIGS & SCI. NOTATION

Evaluate $(8 \times 10^{9}) \div (4 \times 10^{-2})$ in scientific notation.

4 INDEX LAWS

Simplify $(2x^3)^4 \div 8x^5$

5 ZERO & NEGATIVE INDICES

Write 0.0001 as a power of 10.

6 SIMPLIFY & SUBSTITUTE

If $x = -\frac{1}{2}$, find the value of $4x^2 - 2x + 1$.

7 EXPAND ★

Expand $(3x - 2)^2$

8 FACTORISE ★

Factorise $x^2 - 2x - 15$

9 EQUATIONS ★

Solve $x^2 - 5x + 6 = 0$

10 FORMULAS & WORD PROBLEMS

Hiring a hall costs $C = 200 + 15n$ dollars for n guests. How many guests can come for \$1100?

11 PYTHAGORAS

An isosceles triangle has equal sides 13 cm and a base of 10 cm. Find its perpendicular height.

12 TRIGONOMETRY ★

A boat sails 10 km due east, then 10 km due south. Find the true bearing of the boat from its start.

13 AREA & SURFACE AREA ★

Find the surface area of a sphere with radius 3 cm, in terms of π.

14 VOLUME

A cylindrical hole of radius 1 cm is drilled right through a cube with edges 4 cm long. Find the volume left, in terms of π.

15 GRADIENT & MIDPOINT

The interval from $(-3,\ k)$ to $(5,\ 7)$ has gradient $\frac{1}{2}$. Find k.

16 STRAIGHT LINES ★

Find the point where the lines $2x + y = 10$ and $y = x + 1$ meet.

17 DATA

The mean of 6 scores is 15. A seventh score is added and the mean becomes 16. Find the seventh score.

18 PROBABILITY

A box holds 4 red and 6 blue pens. Two are taken without replacement. Find $P(\text{both red})$.

19 EARNING MONEY

Zoe earns \$1200 a week and pays \$180 tax. What percentage of her pay is tax?

20 PROBLEM SOLVING

A tank 2 m long, 1 m wide and 1.2 m high is filled at 40 L per minute. How long does it take to fill?

BONUS The product of two consecutive whole numbers is 132. Find them.

Hints & formulas

Number

Increase by 12%: $\times 1.12$ · decrease by 35%: $\times 0.65$

Percentage change $= \dfrac{\text{change}}{\text{original}} \times 100\%$

Significant figures start at the first non-zero digit

Scientific notation: $a \times 10^n$ with $1 \leq a < 10$

Indices

$a^m \times a^n = a^{m+n}$ · $a^m \div a^n = a^{m-n}$

$(a^m)^n = a^{mn}$ · $(ab)^n = a^n b^n$ · $a^0 = 1$

$a^{-n} = \dfrac{1}{a^n}$, e.g. $2^{-3} = \frac{1}{8}$

Algebra

Fractions: use a common denominator, e.g. $\frac{2a}{3} = \frac{4a}{6}$

$(a+b)(c+d) = ac + ad + bc + bd$

★ $(a-b)^2 = a^2 - 2ab + b^2$

★ $x^2 + bx + c = (x+p)(x+q)$ where $pq = c$, $p+q = b$

Equations & formulas

Expand brackets, collect the unknowns on one side, then solve

$x^2 = 64 \Rightarrow x = \pm 8$

★ Change the subject: undo operations in reverse order

★ $(x-2)(x-3) = 0 \Rightarrow x = 2$ or $x = 3$

Measurement & trig

$c^2 = a^2 + b^2$ · $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$

Surface area: add the areas of all faces

Closed cylinder $SA = 2\pi r^2 + 2\pi rh$ · $V = \pi r^2 h$

★ Cone $V = \frac{1}{3}\pi r^2 h$ · sphere $SA = 4\pi r^2$

★ Elevation: measured up from the horizontal; bearings clockwise from north

Coordinate geometry

$m = \dfrac{y_2 - y_1}{x_2 - x_1}$ · midpoint $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

Length: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

$y = mx + c$: m = gradient, c = y-intercept

★ Perpendicular lines: $m_1 \times m_2 = -1$

Data & chance

Mean = total ÷ count · median: middle value in order

Range = highest − lowest

Independent events: $P(A \text{ and } B) = P(A) \times P(B)$

Without replacement: the second fraction changes

★ Five-number summary: min, Q_1, median, Q_3, max

Money & context

Weekly pay = salary ÷ 52 · fortnightly = ÷ 26

Overtime: time-and-a-half $= 1.5\times$ rate

Commission: a percentage of sales

★ Compound interest: $\times(1+r)$ each year; depreciation: $\times(1-r)$

Answers

1 Week 1 answers

1 -24 $-12 + (-12)$

2 **\$280** 12% of 250 = 30

3 0.053 first significant figure is 5; next digit 3 rounds down

4 a^8 add the indices

5 $1\frac{1}{3}$ $1 + \frac{1}{3}$

6 $2x - 4y$ $4x - 2x$ and $-7y + 3y$

7 $10x - 15$ $6x - 15 + 4x$

8 $3a(3a - 2)$ HCF = $3a$

9 $x = 7$ $2x = 14$

10 $s = 28$ $12 + \frac{1}{2} \times 2 \times 16 = 12 + 16$

11 **25 cm** $c^2 = 49 + 576 = 625$

12 $\sin\theta = \frac{3}{5}$ $\frac{\text{opposite}}{\text{hypotenuse}} = \frac{6}{10}$

13 **150 cm²** 6 faces: 6×25

14 80π **cm³** $\pi \times 16 \times 5$

15 $(5,\ 3)$ $\left(\frac{2+8}{2}, \frac{7+(-1)}{2}\right)$

16 $m = -2, c = 5$ compare with $y = mx + c$

17 14 $70 \div 5$

18 $\frac{3}{4}$ HH, HT, TH out of 4

19 **\$980** 35×28

20 **\$57** $45 + 120 \times 0.10$

B **24, 26 and 28** $n + (n+2) + (n+4) = 78,\ 3n = 72$

2 Week 2 answers

1 $\frac{2}{5}$ $\frac{18}{30} - \frac{1}{5} = \frac{3}{5} - \frac{1}{5}$

2 **\$78** 35% of 120 = 42

3 4.72×10^6 move the decimal point 6 places

4 $4m^6$ $2^2 \times m^{3\times2}$

5 0.001 $\frac{1}{1000}$

6 28 $4 - 4(-2)(3) = 4 + 24$

7 $x^2 + 4x - 12$ $x^2 - 2x + 6x - 12$

8 $7y(2x + 3)$ HCF = $7y$

9 $x = 4$ $5x + 10 = 3x + 18,\ 2x = 8$

10 **7** $b + 4 + 3b + 4 = 36,\ 4b = 28$

11 **15 cm** $x^2 = 289 - 64 = 225$

12 $x \approx 14$ **cm** $\tan 35^\circ = \frac{x}{20},\ x = 20 \times 0.70$

13 9π **cm²** $\frac{1}{4} \times \pi \times 36$

14 **360 cm³** $A = 24;\ V = 24 \times 15$

15 -3 $\frac{-8-4}{3-(-1)} = \frac{-12}{4}$

16 $y = 3x - 4$ y-intercept is -4

17 32 10 values; middle two are 32 and 32

18 $\frac{1}{2}$ 1, 2, 3, 4, 6, 12: $\frac{6}{12}$

19 **\$1320** $38 \times 30 + 4 \times 45 = 1140 + 180$

20 **3.5 km** $14 \times 25\,000 = 350\,000$ cm

B $n^2 + 2n + 1 - (n^2 - 2n + 1) = 4n$ the n^2 and 1 terms cancel

Answers

3 Week 3 answers

1 $1\frac{11}{12}$ $\frac{45}{12} - \frac{22}{12} = \frac{23}{12}$

2 12.5% $\frac{45}{360} = \frac{1}{8}$

3 $0.000\,306$ move the decimal point 4 places left

4 $3x^5$ $12 \div 4$; subtract the indices

5 2 $\frac{1}{4} \times 8$

6 $\frac{5a}{6}$ $\frac{4a}{6} + \frac{a}{6}$

7 $2x^2 + 9x + 4$ $2x^2 + 8x + x + 4$

8 $(x + 3)(x + 4)$ $3 \times 4 = 12$, $3 + 4 = 7$

9 $x = 8$ $2x - 1 = 15$

10 $C = 15$ $27 = \frac{9C}{5}$, $9C = 135$

11 **15 cm** $d^2 = 81 + 144 = 225$

12 $x \approx 9.6$ **cm** $\sin 40° = \frac{x}{15}$, $x = 15 \times 0.64$

13 **108 cm²** $2(24 + 18 + 12)$

14 **3000 L** $2 \times 1.5 \times 1 = 3\ \text{m}^3$

15 **10 units** $\sqrt{6^2 + 8^2} = \sqrt{100}$

16 **Yes** $-3(-2) + 1 = 7$ ✓

17 **favourite beach** the others are numerical

18 $\frac{9}{25}$ $\frac{3}{5} \times \frac{3}{5}$

19 **$13 000** $1\% = 6500$

20 **2 h 5 min** $1500 \div 12 = 125$ min

B **8 cm by 6 cm** $l + w = 14$ and $lw = 48$

4 Week 4 answers

1 -31 $-27 - 4$

2 **$1210** $1000 \times 1.1 = 1100$; 1100×1.1

3 6×10^7 3×2; 10^{4+3}

4 $27a^6b^3$ cube each factor

5 7^{-2} $\frac{1}{7^2}$

6 $\frac{x}{4}$ $\frac{3x}{4} - \frac{2x}{4}$

7 $x^2 - 10x + 25$ $(x - 5)(x - 5)$

8 $5m(4 - m)$ HCF = $5m$

9 $x = 8$ **or** $x = -8$ $\pm\sqrt{64}$

10 **20 km** $2.2d = 44$

11 **12 m** $h^2 = 169 - 25 = 144$

12 **25.2 m** $h = 30 \times 0.84$

13 **48π cm²** $2\pi(9) + 2\pi(3)(5) = 18\pi + 30\pi$

14 **30π cm³** $\frac{1}{3} \times \pi \times 9 \times 10$

15 $B(7,\ 5)$ x: $1 \to 4 \to 7$; y: $-3 \to 1 \to 5$

16 $y = 3x + 2$ $m = \frac{11-2}{3-0} = 3$, $c = 2$

17 **range 9, mode 18** range $21 - 12$

18 $\frac{1}{12}$ $\frac{1}{2} \times \frac{1}{6}$

19 **$3000** $78\,000 \div 26$

20 **90 km/h** $150 \div 1\frac{2}{3} = 150 \times \frac{3}{5}$

B 3 $x = 5$, $y = -2$

Answers

5 Week 5 answers

1 $1\frac{1}{2}$ $\frac{4}{5} \times \frac{15}{8} = \frac{60}{40}$

2 15% $\frac{12}{80} \times 100\%$

3 7.05 the 4th figure is 9, so round up

4 $3x^2y$ $\frac{6}{2}$; subtract the indices

5 $\frac{1}{2}$ $\frac{1}{4} + \frac{1}{4}$

6 -13 $3 - 16$

7 $x^2 + 2x + 4$ $x^2 + 5x + 4 - 3x$

8 $-4a(2 + 3b)$ take out $-4a$

9 $x = 19$ $3x - 12 - 2x - 2 = 5$, $x - 14 = 5$

10 $r = \dfrac{C}{2\pi}$ divide both sides by 2π

11 **Yes** $81 + 1600 = 1681 = 41^2$

12 $\frac{5}{13}$ opposite 5, adjacent 12, so hypotenuse 13

13 **132 cm²** $2 \times 6 + (3 + 4 + 5) \times 10 = 12 + 120$

14 **3 cm** $\pi r^2 \times 8 = 72\pi$, $r^2 = 9$

15 **Yes** both gradients are $\frac{4}{2} = 2$

16 $-\frac{1}{4}$ product of the gradients is -1

17 **3, 5, 8, 11, 14** min, Q_1, median, Q_3, max

18 $\frac{1}{6}$ 6 of the 36 outcomes

19 **400** $480 \div 1.20$

20 **$19 200** $30\,000 \times 0.8 \times 0.8$

B **50 cm²** $s^2 + s^2 = 100$, so $s^2 = 50$

6 Week 6 answers

1 $2\frac{1}{6}$ $-\frac{14}{6} + \frac{27}{6} = \frac{13}{6}$

2 **$400** $460 \div 1.15$

3 2×10^{11} $8 \div 4$; $10^{9-(-2)}$

4 $2x^7$ $16x^{12} \div 8x^5$

5 10^{-4} $\frac{1}{10\,000}$

6 3 $1 + 1 + 1$

7 $9x^2 - 12x + 4$ $(3x)^2 - 2(3x)(2) + 2^2$

8 $(x - 5)(x + 3)$ $-5 \times 3 = -15$, $-5 + 3 = -2$

9 $x = 2$ **or** $x = 3$ $(x - 2)(x - 3) = 0$

10 **60** $15n = 900$

11 **12 cm** half the base is 5: $h^2 = 169 - 25$

12 **135°T** south-east: halfway between 090° and 180°

13 **36π cm²** $4\pi r^2 = 4\pi \times 9$

14 **$(64 - 4\pi)$ cm³** $4^3 - \pi \times 1^2 \times 4$

15 $k = 3$ $\frac{7-k}{8} = \frac{1}{2}$

16 $(3,\ 4)$ $2x + x + 1 = 10$, $x = 3$

17 **22** $7 \times 16 - 6 \times 15 = 112 - 90$

18 $\frac{2}{15}$ $\frac{4}{10} \times \frac{3}{9} = \frac{12}{90}$

19 15% $\frac{180}{1200} \times 100\%$

20 **1 hour** $2.4\ \text{m}^3 = 2400$ L; $2400 \div 40 = 60$ min

B **11 and 12** $n(n + 1) = 132$